April 30

Recruiter-Producer Ratio ($T(\theta)$)

$$T(\theta) = \frac{r \cdot s}{q(\theta) - r \cdot s}$$

r: recruiting cost
s: job separation rate
$q(\theta)$: vacancy filling rate

assume Cobb-Douglas
$q(\theta) = \omega \cdot \theta^{-\eta}$

$$\varepsilon(T) = \frac{d \ln T}{d \ln \theta} = \frac{\theta}{T} \cdot \frac{dT}{d\theta} = \frac{\theta}{T} \cdot T'(\theta)$$

elasticity of the recruiter-producer ratio

Recall the properties of log:

$$\begin{cases} \ln(a \times b) = \ln(a) + \ln(b) \\ \ln(a \div b) = \ln(a) - \ln(b) \\ \ln(a)^c = c \cdot \ln a \end{cases}$$

$$\varepsilon(f \cdot g) = \varepsilon(f) + \varepsilon(g)$$

$$\varepsilon(f/g) = \varepsilon(f) - \varepsilon(g)$$

$$\varepsilon(1/f) = \overset{0}{\varepsilon(1)} - \varepsilon(f) = -\varepsilon(f)$$, b/c $\varepsilon(\text{constant}) = 0$

$$\varepsilon(f+g) = \frac{f \cdot \varepsilon(g)}{f+g} + \frac{g \cdot \varepsilon(f)}{f+g}$$

$$\varepsilon(f-g) = \frac{f \cdot \varepsilon(g)}{f-g} - \frac{g \cdot \varepsilon(f)}{f-g}$$

$$\varepsilon(f^{\eta}) = \eta \cdot \varepsilon(f)$$

$$\mathcal{E}(T) = \mathcal{E}\left(\frac{n \cdot s}{q(\theta) - n \cdot s}\right)$$

$$= \cancel{\mathcal{E}(n \cdot s)}^{\,0} - \mathcal{E}(q(\theta) - n s)$$

$$= -\mathcal{E}(q(\theta) - n s)$$

$$= -\frac{q(\theta) \cdot \mathcal{E}(q(\theta))}{q(\theta) - n s} - \frac{n \cdot s}{q(\theta) - n s} \cdot \cancel{\mathcal{E}(n \cdot s)}^{\,0}$$

$$= \frac{-q(\theta)}{q(\theta) - n s} \cdot (-\eta) \cdot 1 = \eta \cdot \frac{q(\theta)}{q(\theta) - n \cdot s} \qquad 1 + T(\theta)$$

$$\mathcal{E}(T) = \underbrace{\eta}_{\text{elasticity of matching function}} [1 + T(\theta)]$$